



Earn₹s
DSA in C++

if-else
↗ loops

Pattern Printing

(Lecture 10)

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Summary Notes

Contents

1. Squares and Rectangles
2. Triangles & Horizontally Flipped Triangles
3. Special Patterns
4. Vertically Flipped Triangles, Pyramids etc

Nesting of Loops

↓

loop ke andar loop

Ques: Print the given pattern

```
* * * * *
* * * * *
* * * * *
```

Handwritten notes: A box containing 3, 2, 1 and a 1 below it.

```
for(int i=1;i<=2;i++){
    for(int j=1;j<=3;j++){
        cout<<"* ";
    }
    cout<<endl;
}
```

Handwritten notes: A box containing 4, 3, 2, 1 and a 1 below it, and another box containing 1 and a j below it.

Output

```
* * *
* * *
```

Star Rectangle

Ques: Print the given pattern

 **Earn₹s**
'n' = 'm'

'Same'

HW

```
* * * *  
* * * *  
* * * *  
* * * *
```

Star Square

Ques: Print the given pattern

1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4

$n = 3$

1 2 3
1 2 3
1 2 3

$n = 4$

Number Square

Ques: Print the given pattern ↗ $n \leq 26$  **Earn ₹s**

$n = 4$

A B C D
A B C D
A B C D
A B C D

$n = 5$

A B C D E
A B C D E
A B C D E
A B C D E
A B C D E

Alphabet Square

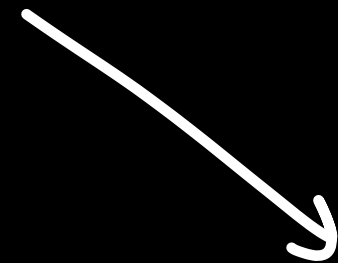
$n=3$

1 2 3

1 2 3

1 2 3

j

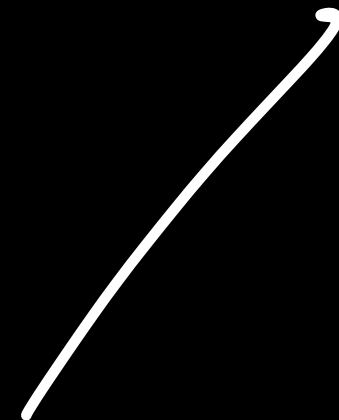


65 66 67

65 66 67

65 66 67

$(j+64)$



A B C

A B C

A B C

$(\text{char})(j+64)$

HW: Print the given pattern

```
a b c d
a b c d
a b c d
a b c d
```

Alphabet Square 2

Ques: Print the given pattern

1 2 3 4 → j

1 1 1 1

2 2 2 2

3 3 3 3

4 4 4 4

↓

i

n = 4

```
for (i → 1 to n) {
    for (j → 1 to n) {
        print(i)
    }
}
```

Number Square 2

HW: Print the given pattern

A A A A
B B B B
C C C C
D D D D

`(char)(i+64)`

↓
outer variable

Alphabet Square 3

HW: Print the given pattern

1 2 3 4 → j
1 a a a a
2 B B B B
3 C C C C
4 D D D D
↓
i
n = 4

odd rows `odd(i)` → small alphabet
else capital

Alphabet Square 4

Ques: Print the given pattern

```

1 *
2 * *
3 * * *
4 * * * *
↓
1

```

$n=4$

$n=3$

```

1 *
2 * *
3 * * *
↓
1

```

$3 \rightarrow j$

$j \leq i$

$n=5$

```

1 *
2 * *
3 * * *
4 * * * *
5 * * * * *
↓
1

```

Star Triangle

Ques: Print the given pattern

```
1
1 2
1 2 3
1 2 3 4
```

Number Triangle

HW: Print the given pattern

A
A B
A B C
A B C D

$n=4$
1
2 2
3 3 3
4 4 4 4

Alphabet Triangle

HW: Print the given pattern

$n = 5$

1 **1**
2 **A B**
3 **1 2 3**
4 **A B C D**
5 **1 2 3 4 5**
↓
i

```
for(i = 1 to n)
|  for(j = 1 to i) {
|  |
|  }
}
```

Alphanumeric Triangle

Method-1

Ques: Print the given pattern

```
1 * * * *
2 * * *
3 * *
4 *
```

$n=4$

```
1 2 3 → j
1 * * *
2 * *
3 *
↓
n=3
↓
```

$$i + j_{\max} = n + 1$$

```
1 2 3 4 5 → j
1 * * * *
2 * * *
3 * *
4 *
5 *
```

$n=5$

$$j \leq n + 1 - i$$

Star Triangle Horizontally Flipped

Method-2



Earn ₹s

Ques: Print the given pattern

4 * * * *

3 * * *

2 * *

1 *

↓
n=4

↓

```
for(i → n to 1){  
    for(j → 1 to i){  
        _____  
    }  
}
```



Star Triangle Horizontally Flipped

HW: Print the given pattern

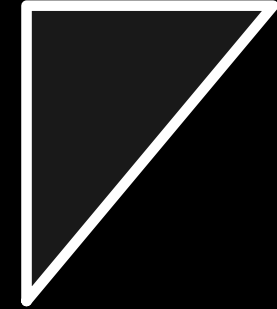
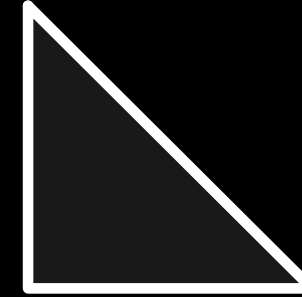
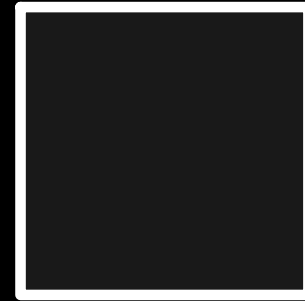
```
1 2 3 4
1 2 3
1 2
1
```

Number Triangle Horizontally Flipped

HW: Print the given pattern

```
a b c d
a b c
a b
a
```

Alphabet Triangle Horizontally Flipped



```
for(i = 1 to n){
  | for(j = 1 to n){
  | | —————
  | }
}
```

```
for(i = 1 to n){
  | for(j = 1 to n+1-i){
  | | —————
  | }
}
```

1	2	3	...	A	B	C	...
1	2	3	...	A	B	C	...
:				:			
1	2	3	...				

j

(char)(j+64)

1	1	1	...
2	2	...	
:			
n	n	n	...

i



Ques: Print the given pattern ($m \times n$)

1 2 3 4 5 6 7 $\rightarrow j$

(Hint : If-Else)

1 * * * * *

2 * # # # # *

3 * # # # # *

4 * # # # # *

5 * * * * *

↓

6

Hollow Rectangle

Ques: Print the given pattern 'n' → odd

1 2 3 4 5 → j

1 — — * — —
 2 — — * — —
 3 * * * * *
 4 — — * — —
 5 — — * — —
 ↓
 i

n = 5

n = 3

— x —
 x x x
 — x —

(hint · square)
 if-else

$$\text{mid} = \frac{n+1}{2}$$

if (i == mid || j == mid)

HW: Print the given pattern $n \rightarrow \text{odd}$



Earn₹s

	1	2	3	4	5
1	*	—	—	—	*
2	—	*	—	*	—
3	—	—	*	—	—
4	—	*	—	*	—
5	*	—	—	—	*

Star Cross

Ques: Print the given pattern

1 2 3 4 $\rightarrow j$

(Hint . Extra Variable)

1 **1**
2 **2 3**
3 **4 5 6**
4 **7 8 9 10**

$\int n=4$
 i

Floyd's Triangle

Ques: Print the given pattern

1 2 3 4

1
2 1 3
3 1 3 5
4 1 3 5 7

$n=4$

In each line you need to
print first 'i' odd numbers

Odd Number Triangle

Ques: Print the given pattern

```
1
01
101
0101
```

$n=4$

Try karna.

Binary Triangle